

Reassessing the Effect of Bank Branching Deregulation on Income Inequality

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Abstract

Published in the *Journal of Finance* in 2010, “Big Bad Banks? The Winners and Losers from Bank Deregulation in the United States” is a well-cited paper finding that bank branching deregulations staggered across various U.S. states significantly reduced income inequality within these states. Their design, which focuses on staggered DiD and event study specifications with two-way fixed effects (TWFE) is the subject of scepticism, given more recent econometric findings that find such designs can create bias through heterogeneous treatment effects and comparisons of treated subjects on already-treated subjects. This paper seeks to assess such concerns by implementing modern DiD estimators (Callaway-Sant’Anna, Borusyak & Jaravel & Spiess) on the original analysis done by Beck & Levine & Levkov (2010). With these new estimators, I find mixed results in favour and against Beck et al. Though both estimators show statistically insignificant or even worsening effects on income inequality, I show that both estimators also suffer from parallel trends violations and sensitivity concerns in the context of Beck et al. sample period that lend some evidence in favour of the negative inequality effect observed in Beck et al., though with the estimates being more attenuated than in the original TWFE design.

1 Synopsis

While modern financial and banking regulations are often centred around questions of economic stability, one important policy concern in regards to these policies is their effect on income inequality. Large consolidated financial institutions have been historically treated with suspicion; with the public sentiment prevailing that these large institutions primarily serve the interest of wealthy financiers and industrialists. This sentiment has grown further in popularity since the financial crisis of 2008, with the historic bailouts of the financial industry and the “too big to fail” mantra becoming rallying cries for progressive policy makers on the unequal treatment of banks and the need for greater control over a perceived banking oligopoly. In contrast, opponents argue that banking regulations primarily protect established large banks, creating a higher cost of entry that restricts banking competition and hurts the poor. This raises the question: do banking regulations have an impact on income inequality, and if so through what channel?

This is the question that Beck & Levine & Levkov (2010) seek to study. In doing so, the authors face a variety of challenges. Choices on bank regulations and de-regulations are not exogenous: passed into law by legislatures under particular political and economic characteristics that also affect income inequality. If, for example, branch deregulation coincides with various other economic policy regimes in ways that affect the income distribution, then a comparison of pre- and post-deregulation inequality would confound the policy effect with the underlying economic trend. Furthermore, previous research has shown that high levels of inequality can also affect political decisions, like banking laws, through mechanisms like lower voter turnout, greater lobbying, or support for particular political parties which creates for reverse causality (Song & Kang, 2025; López & Dubrow, 2020). To tackle the issue of economic and political endogeneity, Beck et al. exploit exogenous variation in a set of time-staggered state-wide banking deregulations regarding bank branching. Historically, many states had historical banking regulations preventing banks from opening multiple branches within a state. Between mid 70s to late 90s, these regulations were gradually repealed until 1994 with the passing of the Riegle-Neal Interstate Banking and Branching Efficiency Act, which repealed intrastate branching restrictions on a national level.

1.1 Methodology

Using onset of deregulation as a treatment, Beck et al. employ both a staggered differences-in-differences (DiD) design and an event study design to compare income inequality between states who had implemented branching deregulation and those who didn’t before and after the deregulation went into law. The staggered DiD is of the following form:

$$Y_{st} = \alpha + \beta D_{st} + \delta X_{st} + A_s + B_t + \epsilon_{st} \quad (1)$$

For each state s in year t , Y_{st} indicates a measure of income inequality in that state. For this measure, the authors use three separate measurements: the Gini coefficient, which captures the deviation of the income distribution from a perfectly uniform income distribution; the Theil index, which though similar to the Gini is decomposable into within group and between group inequality; and the log difference between the 90th

(75th) and 10th (25th) percentile incomes to study impacts on particular areas of the distribution. D_{st} is an indicator for whether a state s had deregulated bank branching in year t . X_{st} is a set of control for time-varying state level variables, including the unemployment rate, the growth rate per capita Gross State Product (GSP), and various socio-demographic characteristics such as the proportion of African Americans and high school drop-outs. Finally, to account for time-invariant state characteristics (e.g., geography, institutional history, political culture, industrial composition, etc.) and nation-wide economic shocks (e.g., business cycles, federal policy changes, etc.) that affect income inequality, Beck et al. include A_s and B_t as state and year fixed effects in a TWFE model. The coefficient of interest is β , which is an average treatment effect on the treated (ATT) that captures the average change in inequality that occurs from state moving a pre-deregulation to the post-deregulation state, relative to states that were not deregulated or had not deregulated at that time, with standard errors being clustered at the state level.

To study the dynamic effects on income inequality and test for parallel trends in the DiD design, the authors further employ an event study specification of the form,

$$\log(Gini)_{st} = \alpha + \sum_j \beta_j D_{st}^j + A_s + B_t + \epsilon_{st} \quad (2)$$

Where D_{st}^j is an indicator for whether a state is in the the j -th year before or after deregulation.

1.2 Identification Assumptions

For identification, the authors rely on a number of assumptions. The first assumption is parallel trends, or that without deregulation, states that deregulated and those that didn't would've followed similar trends in income inequality, conditional on the control and the TWFEs. Using the event study design, Beck et al. plot out the estimated effects on income inequality for each year pre-deregulation, showing that the pre-trends in log Gini coefficient are not statistically different from zero. The next two assumptions are the no-anticipation assumption, where anticipated future deregulation didn't affect income inequality, and treatment exogeneity, or that the factors affecting the onset of branching deregulations are uncorrelated with income inequality. For identification on both these assumptions, the authors utilize findings in Kroszner & Strahan (1999). Using a hazard model, Kroszner & Strahan (1999) argue that the onset of these branching deregulations were driven by technological change, such as the introduction of the ATM, money market mutual funds, and improved communication technologies. These banking innovations reduced the benefit of the local monopoly power banks held under this regulation, and subsequently reduced the political incentive to lobby for such legislation. Given these results, Beck et al. argue that this technological determinant in banking deregulation timing was both plausibly unanticipated, given the slow movement of force driving these deregulations, and is a political channel that is independent of income inequality. Beck et al. further defend this claim by plotting pre-deregulation Gini coefficients (Table 1) to show no correlation of prior income inequality with branch deregulation. Finally, there is a stable-unit-treatment-value-assumption, or SUTVA, which in this case means that the onset of branching deregulation doesn't affect inequality in other states. One primary concern of Beck et al. is inter-state migration, or that branching deregulations affects how people migrate

into or out of the deregulating state. The authors thus address this concern by testing whether branching deregulation affected migration outflows from a state, finding no such effect.

1.3 Data

For their data, the author use individual-level data from the Current Population Survey (CPS) to construct inequality measures, with the data restricted to individuals 25 to 54 with nonnegative personal income between 1976-2007. Adding both deregulation date data from Jayaratne & Strahan (1996) and control data from the Bureau of Economic Analysis, Bureau of Labour Statistics, and Kroszner & Strahan, the final sample data consists of 48 states plus the District of Columbia, with both South Dakota and Delaware being dropped due to their notably different banking structures with highly concentrated credit card operations. In this sample period, 37 states deregulated during this time, leaving 12 states that had already deregulated before 1976.

1.4 Findings

Using the above specifications, Beck et al. find that these set of banking deregulations reduced state-wide income inequality. Using a percentile-by-percentile analysis (Figure 2), the authors find that this income inequality reduction is driven by increased income shares for the 5-35th percentile, with higher percentile groups experiencing no statistically significant change in income share. Further dynamic results show this change to be persistent, with income inequality gradually decreasing until convergence at around 3.5% lower Gini coefficient.

To study heterogeneity of treatment, the authors first perform a set of heterogeneity analyses where deregulation timing is interacted with pre-deregulation state characteristics. Using this, Beck et al. find that the effects were larger in “unit banking” states, which had the most restrictive branching regulations pre-treatment, along with states with less population density, larger asset shares from small banks, and a higher share of small firms. Next, using the Theil index decomposed into between-group and within-group components to hypothesize on various mechanisms. Notably, their results indicate that inequality reductions we’re driven by declining inequality among paid workers, with inequality among the self-employed and inequality between the self-employed and paid workers seeing no significant change. Using further decompositions on education level, Beck et al. find that much of the change in concentrated in declining inequalities among less-educated workers. Given a further decomposition on age, finding no statistically significant differences in inequality reduction across age groups, Beck et al. conclude that these results are neither driven through increased entrepreneurship (given the lack of effect on self-employed), or through increased education among the poor (given a lack of difference in effect for younger individuals), but is instead likely result of a labour demand shift. More specifically, the authors argue that increased banking competition relaxed credit constraints for firms, who respond by increasing their output and demand for lower-skilled workers. These results are reflected in both increases in wages for unskilled workers, increases in relative hours worked, and decreases in unemployment following the banking deregulations.

2 Methodological Innovations and Relevance

Since Beck et al. was published in 2010, there has been numerous econometric findings regarding DiD research designs with TWFE. In light of these findings, I present the following concerns with the model specification Beck et al. and how I plan to diagnose these problems.

2.1 Heterogeneity Bias & Unequal Weighting

Goodman-Bacon (2021) shows that under a TWFE model with staggered treatment, coefficients of the model are derived from a weighted average of all possible two-by-two group and period DiD comparisons that are constructable from the data. This includes treated versus never-treated, and treated versus not-yet-treated comparisons, but also includes treated versus already-treated comparisons.

In relation to Beck et al., there exists no “never-treated” group of states as all states pass branching deregulations by the 1994 passing of Riegle-Neal. As such, if for any two states we define a “pre” period before either state is deregulated, a “post” period where both are deregulated, and a “mid” period where one state is deregulated and one is not, the staggered DiD estimate can be decomposed into the following:

$$\hat{\beta}_{t_1 t_2}^{2 \times 2} = \left(\bar{Y}_{t_1}^{mid(t_1, t_2)} - \bar{Y}_{t_1}^{pre(t_1)} \right) - \left(\bar{Y}_{t_2}^{mid(t_1, t_2)} - \bar{Y}_{t_2}^{pre(t_1)} \right) \quad (3)$$

$$\hat{\beta}_{t_2 t_1}^{2 \times 2} = \left(\bar{Y}_{t_2}^{post(t_2)} - \bar{Y}_{t_2}^{mid(t_1, t_2)} \right) - \left(\bar{Y}_{t_1}^{post(t_2)} - \bar{Y}_{t_1}^{mid(t_1, t_2)} \right) \quad (4)$$

Where t_1 and t_2 are the years of deregulation for two particular states being compared, with $t_2 > t_1$. Thus, the DiD estimator is the following:

$$\hat{\beta}^{DD} = \sum_{t_2} \sum_{t_2 > t_1} s_{t_1 t_2} \left[\mu_{t_1 t_2} \hat{\beta}_{t_1 t_2}^{2 \times 2, t_1} + (1 - \mu_{t_1 t_2}) \hat{\beta}_{t_2 t_1}^{2 \times 2, t_2} \right] \quad (5)$$

Where the estimate is a weighted average of the DiD comparisons between states that have deregulated and those that haven’t deregulated yet ($\hat{\beta}_{t_1 t_2}^{2 \times 2}$), and between states that have deregulated and those that already previously deregulated ($\hat{\beta}_{t_2 t_1}^{2 \times 2}$). Defining a particular ATT for each deregulation year group,

$$ATT_{t_1}(\tau) = E[Y_{st}^1 - Y_{st}^0 | t_1, t = \tau] \quad (6)$$

We derive the following for each decomposed estimate:

$$\hat{\beta}_{t_1 t_2}^{2 \times 2} = ATT_{t_1}(Mid) + \Delta Y_{t_1}^0(Mid, Pre) - \Delta Y_{t_2}^0(Mid, Pre) \quad (7)$$

$$\hat{\beta}_{t_2 t_1}^{2 \times 2} = ATT_{t_2, Post(t_2)} + \Delta Y_{t_2}^0(Post(t_2), Mid) - \Delta Y_{t_1}^0(Post(t_2), Mid) - (ATT_{t_1}(Post) - ATT_{t_1}(Mid)) \quad (8)$$

The terms $\Delta Y_{t_2}^0(Mid, Pre) - \Delta Y_{t_1}^0(Mid, Pre)$ in (7) and $\Delta Y_{t_2}^0(Post(t_2), Mid) - \Delta Y_{t_1}^0(Post(t_2), Mid)$ in (8) represent bias if parallel trends is violated. Assuming the original parallel trends argument in Beck et al. holds, then for (8) we are still left with $(ATT_{t_1}(Post) - ATT_{t_1}(Mid))$, which reflects an additional bias resulting from heterogeneous treatment effects. As previously discussed, Beck et al. show that changes in inequality are heterogeneous to various state characteristics like population density and proportions of small banks

and firms, which implies this bias is different from 0 and is problematic for the results. Thus, given that the DD estimated weight is,

$$\mu_{t_1 t_2} = \frac{1 - D_{t_1}}{1 - (D_{t_1} - D_{t_2})} \quad (9)$$

where D_{t_1} is the fraction of time periods group t_1 spends treated, then the states who get treated early on (mid to late 70s deregulation) receive weight close to zero on the good comparison and weight close to one on the bad comparison, which implies,

$$\hat{\beta}_{t_2 \text{ early}}^{DD} \approx \sum_{t_2} \sum_{t_2 > t_1} s_{t_1 t_2} \left[\hat{\beta}_{t_2 t_1}^{2 \times 2, t_2} \right] \quad (10)$$

Where the weights fall disproportionately on the bad comparisons.

To compound the issues discussed above, we also have the 12 other states that had deregulated branching before the sample period. These states are included in the original specification under the assumption that they don't effect the estimate β . However, we now know this is not the case, as these states join the “already-treated” group. Thus, in any comparison with these group of states, which we'll denote t_0 , it holds that $D_{t_0} = 1$ as they are treated across the whole sample period. The result implies that Beck et al. contains a great number of two-by-two DiD comparisons which are heterogeneously bias and weigh our DiD estimate away from the true parameter.

To diagnose this problem, I turn to using Goodman-Bacon Decomposition. This decomposition explicitly derives each two-by-two comparison groups (3) and (4), and derives the weight that each group receives in the DiD estimator. Running this diagnostic on Beck et al., we would expect the diagnostic to pass if the vast majority of the weight derives from clean comparisons between deregulating states and states that have yet to deregulate, with these comparison estimates being similar to the original TWFE coefficient. In contrast, if the Goodman-Bacon decomposition fails, then we would observe that substantial weight in the TWFE model falls on these bad comparisons, either involving the 12 pre-sample deregulated states or on already deregulated states being compared with more recent deregulated states, with this weight distorting the group estimates from the original TWFE coefficients. This would imply that the the estimated coefficients in Beck et al. are largely being driven by these bad comparisons with the ATT no longer identified.

2.2 Heterogeneity-Robust Differences-in-Differences Estimators

In light of the negative weighting problem discussed above, numerous new “robust” DiD estimators have been constructed to mitigate this potential heterogeneity bias in staggered TWFE designs like Beck et al. The two most prominent estimators are from Callaway & Sant'Anna (2021) and Borusyak & Jaravel & Spiess (2024) I go into detail for each of the following estimators, and their relevance to Beck et al.

2.2.1 Callaway & Sant'Anna (CSA)

In CSA, rather than estimating one ATT, a set of ATTs are estimated for each group of states deregulating in time period g at the time of t . In estimating this set of ATTs,

a base period of $g-1$ is created to measure the effect across treatment change, and each comparison of group g is chosen such that the control group is one that hasn't yet been treated. As a result, CSA removes any DiD comparisons arising from already-treated units and only keeps the good treated versus non-treated comparisons that DiD is meant to estimate. Once all these group-time ATTs are estimated, they can then be aggregated to produce an overall ATT. In regards to Beck et al., this would mean that all comparisons are driven between groups states that deregulate in one year, and states that haven't yet deregulated by that year. For the states that deregulated before 1976, given these states have no non-treatment phase, these states would be restricted from our estimation entirely, which reduces our sample size down for 49 to 37 states.

One important element in CSA is the choice of control group and it's relation to the parallel trends assumption. Callaway & Sant'Anna show that the identifying assumptions for the DiD changes depending on the type of control group used in a staggered DiD estimator. For a control group of never-treated units, their control status remains stable, so for identification it only has to hold that each treated group shares parallel trends with the never-treated group. On the other hand, if the control group is made of units that eventually get treated, then to ensure identification the control group is no longer stable, shrinking as more units receive treatment to avoid bad comparisons. In relation to Beck et al., this is important as there are no states that never deregulate. Thus, the entire control group consists of states yet to deregulate by the time of the ATT estimate of deregulated states g at time t , which is a group that grows smaller into the later estimates. One concern from this is that for the states that deregulate the latest (Arkansas, Iowa, Minnesota, etc.), Kroszner & Strahan argue that these states had the strongest political opposition to deregulation. As such, these states may systematically differ from early deregulating states that are removed from the control group, creating concerns about parallel trends holding among these later deregulating states.

To test for this concern, I run sensitivity tests, where different CSA estimates are created by re-estimating CSA over two separate sample windows: 1975–1993 and 1975–1990. In each window, cohorts that deregulated after the window endpoint are reassigned to the not-yet-treated control group, and observations after the endpoint are dropped from the panel. I also exclude the 1994 Riegle-Neal-induced deregulations (Iowa, Arkansas, Minnesota) from the 1975–1993 window, given these states differ substantially from our sample in that they did not voluntarily deregulate branching. If this diagnostic were to pass, then we would observe similar CSA estimates from the restricted windows to the full sample CSA estimate, implying that characteristics of late-deregulating states do not affect the CSA estimates. By contrast, a failure of this test would imply that our window-restricted CSA estimates differ substantially from the original CSA in both sign and magnitude. This would then raise concerns about the identification assumption holding in late-deregulating states.

2.2.2 Borusyak & Jaravel & Spiess (BJS)

BJS is an estimator that attempts to solve the same heterogeneity bias problem that CSA addresses. The difference is that while CSA directly compares treated to untreated groups, BJS attempts to estimate a counterfactual untreated outcome for

treated units, essentially asking what each treated unit would look like had it not been treated. To do this, BJS first estimates a TWFE model on the untreated units to estimate untreated outcomes. In relation to Beck et al. the model would look like,

$$Y_{st} = A_s + B_t + \delta X_{st} + \varepsilon_{st}, \quad \forall (s, t) \text{ s.t. } D_{st} = 0 \quad (11)$$

where A_s and B_t are the state and year fixed effects and X_{st} are the covariates mentioned before. This model is subsequently run on all observations of states who hadn't yet been deregulated by the time of the observation, which also excludes all the states who deregulated before 1976. Once this done, we then use these results to estimate potential un-deregulated outcomes for every deregulated observation,

$$\hat{Y}_{st}^0 = \hat{A}_s + \hat{B}_t + \hat{\delta} X_{st} \quad \text{for all } (s, t) \text{ where } D_{st} = 1 \quad (12)$$

which provides an estimated treatment effect of $Y_{st} - \hat{Y}_{st}^0$.

Compared to CSA, BJS is more efficient as the control group is stable and consists of all untreated observations, creating for tighter standard errors. However, one notable concern for BJS in the context of Beck et al. is the differences in identification assumptions. As mentioned in CSA, each group-time ATT is constructed using a base period of $g - 1$ to estimate the change in inequality compared to a control group. This means that for CSA, parallel trends only has to hold for adjacent short pre- g periods at each comparison, which is fairly flexible and reasonable under the original parallel trends assumption in Beck et al. Because BJS constructs potential outcomes using many pre-treatment years, the parallel trends assumption is much stronger, requiring that parallel trends hold across the untreated panel. This is potentially problematic for Beck et al. given the vast economic changes that occurred over the 1976-2007 data sample that had heterogeneous effects on states and their levels of income inequality (e.g, the decline of manufacturing). For states who were late deregulators, their long pre-treatment data may capture some of these long-run differential effects in our potential outcomes model, which again has the potential to bias the estimated treatment effect.

To test this, I estimate both CSA and BJS and compare their effects. For this test to pass, we would ideally like to see similar estimates for CS and BJS, implying that differences in the parallel trends assumption is not affecting our results. If on the contrary we get vastly different estimates for CSA and BJS, this would likely suggest that parallel trends is violated for distant pre-deregulation periods. Either way, given the robustness of CSA to violations of distant pre-trends, we treat CSA as our primary modernized estimate going forward.

2.3 Pre-Trends Testing Beyond the Event Study

In Beck et al., the main argument used supporting the parallel trends assumption is the insignificant pre-trends they observe in the event study. As explained in Ram-bachan & Roth (2023), this is rather problematic, as a pre-trend test of this type has low statistical power. More specifically, for each lag variable in the event study, there is an increasingly smaller sample to derive two-by-two comparisons. Thus, when Beck et al. present lag estimates for up to 10 years pre-deregulation, there are fewer state observations that have data 10 years before deregulation, which lowers the sample size

and increase the size of the confidence intervals as observed in figure 3. This decreasing comparison size affects the power of these pre-trends test, and creates room for suspicion that economically meaningful pre-trends exist but are not detected in this test.

Therefore, to diagnose this I turn to using the “Honest DiD” estimates proposed in Rambachan & Roth (2023). These estimates create confidence intervals that are honest even if parallel trends are violated. Essentially, this asks the question of what the range of treatment effects is, given violations of parallel trends that would not have been detected under regular specifications? A pass of this diagnostic would mean that the treatment effect measured post de-regulation would still be statistically significant, implying that even under parallel trends violations the estimates are still robust. In contrast, if the treatment effect is no longer statistically significant, then there is plausible concern that coefficients pre-deregulation could explain the estimated treatment effect on income inequality.

3 Diagnostics & Results

To diagnose the heterogeneity bias and weighting, on figure 1 I plot the estimates of the two-by-two comparisons derived from the Goodman-Bacon Decomposition by their weight in the DiD estimator. From the figure, it’s observed that all of the negative DiD effect on income inequality that was measured on Beck et al. is coming through these heterogeneously biased comparisons: either later deregulating states compared with earlier deregulating states, or deregulating states compared with the 12 always-deregulated states. Furthermore, almost all these bad comparisons are given equal weight to the good treated-vs.-untreated comparisons, with some bad comparisons being given the most weight of any comparisons comprising the DiD estimator. As a result, this indicates a resounding failure of Beck et al. in our Goodman-Bacon Diagnostic, suggesting that the measured effects on income inequality in the paper are likely the product of biased comparisons.

In light of these findings, we turn attention to estimating both CSA and BJS estimators on the original dataset in Beck et al. to visualize the magnitude by which the original estimates in the paper is being driven by a biased TWFE design. On table 1, I replicate the effects of state branching deregulation on income inequality from both the original paper (Table II, Panel A), and present them alongside results from the two robust estimators across all income inequality measures used. Here, the robust estimators raise some initial concerns in Beck et al. Looking at the estimates from the CSA on Panel B, we observe no statistically significant effects of branching deregulation on wealth inequality relative to the Gini or the Theil index, with inequality seemingly appears to increase at a statistically significant level relative to the 75th/25th percentile gap. With BJS, the difference in results is even more extreme, with BJS estimating that branching deregulation had a statistically significant worsening effect on income inequality across all income inequality measures. On figure 5 I compare the differential effects log income effects by income percentile from figure 2 in Beck et al. and our results under robust CSA estimation. In contrast to Beck et al. where they find a positive income effects on low income earnings, the CSA estimator appears to statistically significant negative effect on log incomes for workers in the lower-middle of the

income distribution, with the magnitude of effect being comparable in magnitude to those observed Beck et al. but now reversed.

At this point, it should be noted that that our BJS evidently differ substantially from our CSA estimates, which would indicate concern with violation of the stronger parallel trends assumptions in BJS that was previously discussed. To further test this concern, on figures 2, 3, and 4 I compare the original event study results from Beck et al. (Figure 3) and compare them again to event studies estimated with CSA and BJS. Similar to table 1, we observe significant differences in event-time effects across the estimators. More notably, we also observe significant differences in our CSA and BJS estimates, with CSA now estimating apparent worsening effects on income inequality over time (though the CIs are quite large), and the BJS estimating a small worsening effect on income inequality that collapses to 8 years. Alongside these differences, there is also observable issues in the pre-trends of the BJS event study, with the pre-trends point estimates being very large and in some cases statistically different from zero. As such, this is evidence in favour of failing our diagnostic on BJS estimator, implying that the long-run parallel trends assumption in BJS is violated, with the potential outcomes model in BJS capturing long-run differential effects in states income inequality trends that biases its treatment effect.

Turning back to CSA, there remains the concern that our CSA estimates may suffer from bias resulting from parallel trends violations in later deregulating states. On table 2, I report results of branching deregulation on the Log Gini from the original TWFE design, dropping the 12 always-treated states, and present our previous CSA estimate alongside CSA estimates from time restricted windows. From table 2, it's first noted that the TWFE estimate remains statistically significant and negative once the 12 always-treated states are dropped, though the effect is smaller. More importantly, our results indicate that our CSA estimates are insensitive to sample period changes, with estimates in the restricted windows being negative (as opposed to the positive original estimate), and even being somewhat statistically significant in the 1975-1993 window. This is a significant finding, as it shows a failure of our CSA sensitivity diagnostic, implying that our original CSA estimate is potentially biased from differential control groups, but also in stark contrast to our other findings provides some evidence in favour of Beck et al. Specifically, now that we've restricted our control group to one that is more comparable in time, we see that our estimates are once again negative. While these results are not as large or as statically significant as in Beck et al., it's feasible that this lack of statistical significance can be pinned down to a lower sampling period.

Our concerns about potential bias in CSA are furthered in table 3, where we present results from the Honest DiD estimates constructed by Rambachan & Roth. The results show that with even modest increases in $\bar{M}$, which represents the bounds on magnitude of post-deregulation parallel trends violations, the estimated increase in inequality observed in parts of CSA lose significance, implying that our CSA estimates are also sensitive in effect to parallel trends violations.

4 Discussion & Conclusion

In this paper, I re-estimated findings in “Big Bad Banks? The Winners and Losers from Bank Deregulation in the United States” by Beck et al. to address concerns regarding internal validity in the staggered TWFE design that was used. In doing so, I find mixed results. On the one hand, both overall ATT estimates from heterogeneously robust estimators like Callaway-Sant’Anna, and Borusyak & Jaravel & Spiess appear to indicate no statistically significant effect, or even a worsening effect of intrastate branching deregulation on income inequality. On the other hand, we’ve given plausible evidence to suggest that both the BJS and CSA estimates are contaminated by short-run and long-run state differential effects that violate their respective parallel trends assumptions. Furthermore, by restricting CSA to a later-term with more comparable control groups, we found that results once again became negative and partially statistically significant, providing some support for the general effect observed in Beck et al. though with more attenuated estimates. This analysis primarily focused on the general estimates in Beck et al. This seems room for further analysis regarding the effects measured on state and industry heterogeneity, and the decomposing of the Theil index effects for mechanism analysis of the paper’s results.

The significance of this finding to Beck et al. is that despite the previously unknown issues in their original staggered TWFE specification, it’s not evidently clear from these robust estimators that the inequality reducing effect observed in their original paper is false. This result is in contrast to that found in Baker & Larcker & Wang (2021), who in a paper addressing the short-comings of staggered TWFE designs in the financial economics literature flag Beck et al. as an example where CSA may be more robust; finding similar statistically insignificant to positive effects that we also initially observed in table 1. However, while the analysis in Baker & Larcker & Wang is valid, their analysis is not exhaustive, merely using Beck et al. as a short illustration of how heterogeneously robust estimators like CSA can drastically change estimates. My results provide more nuance to the results in Baker & Larcker & Wang, in that under sensitivity analysis with CSA we can still observe more similar effects to those in the paper and must be careful in dealing with the new identification assumptions that these staggered DiD estimators require.

5 Tables & Figures

Table 1: The Impact of Deregulation on Income Inequality: TWFE vs. Modern Estimators

	Logistic Gini (1)	Log Gini (2)	Log Theil (3)	Log 90/10 (4)	Log 75/25 (5)
<i>Panel A: Beck et al. (TWFE)</i>					
Bank deregulation	−0.038*** (0.013)	−0.022*** (0.007)	−0.040** (0.016)	−0.134** (0.058)	−0.077*** (0.019)
Observations	1,519	1,519	1,519	1,519	1,519
<i>Panel B: Callaway and Sant’Anna (2021)</i>					
ATT	0.017 (0.013)	0.011 (0.008)	0.022 (0.016)	0.072 (0.069)	0.058*** (0.020)
<i>Panel C: Borusyak, Jaravel, and Spiess (2024)</i>					
ATT	0.032*** (0.010)	0.019*** (0.006)	0.040*** (0.011)	0.160*** (0.045)	0.066*** (0.014)
Observations	828	828	828	828	828

Notes: Panel A reports two-way fixed effects estimates from Beck, Levine, and Levkov (2010), Table II, Panel A. The dependent variables are: (1) logistic transformation of the Gini coefficient, (2) natural logarithm of the Gini coefficient, (3) natural logarithm of the Theil index, (4) natural logarithm of the ratio of the 90th to 10th percentiles, and (5) natural logarithm of the ratio of the 75th to 25th percentiles. All TWFE regressions include state and year fixed effects with standard errors clustered at the state level. Panel B reports the overall average treatment effect on the treated (ATT) from the Callaway and Sant’Anna (2021) estimator using not-yet-treated states as the control group. Panel C reports the overall ATT from the Borusyak, Jaravel, and Spiess (2024) imputation estimator. For Panels B and C, the 12 states that deregulated before the start of the sample period in 1976 are excluded. The BJS sample is further reduced ($N = 828$) because counterfactual outcomes cannot be imputed for treated observations in years when no untreated states remain. Standard errors are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

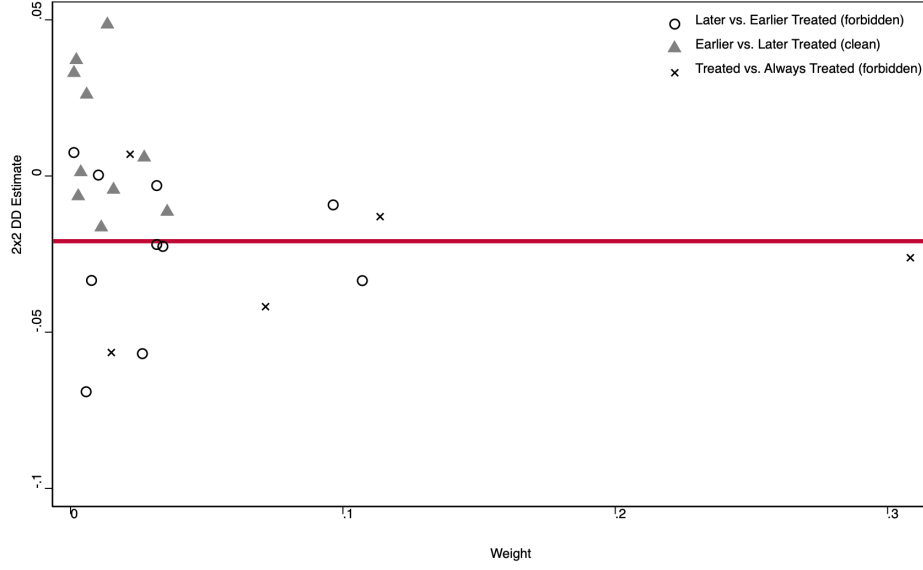


Figure 1: Goodman-Bacon Decomposition

Notes: This figure displays the Goodman-Bacon (2021) decomposition of the two-way fixed effects estimate of bank branching deregulation on the log Gini coefficient. Each point represents one of the 2×2 difference-in-differences comparisons implicitly contained in the TWFE estimator, plotted against the weight it receives in the overall estimate. The horizontal axis reports the weight $s_{k\ell}$ and the vertical axis the corresponding 2×2 DD estimate. Grey triangles denote comparisons in which a later-deregulating state serves as the clean not-yet-treated control for an earlier-deregulating state. Hollow circles denote comparisons in which an already-deregulated state serves as the control for a later-deregulating state. Finally, crosses denote comparisons on one of the twelve states deregulated prior to 1976. The solid red line marks the overall TWFE point estimate of -0.021 . Deregulation years are binned into five-year windows to produce a balanced decomposition.

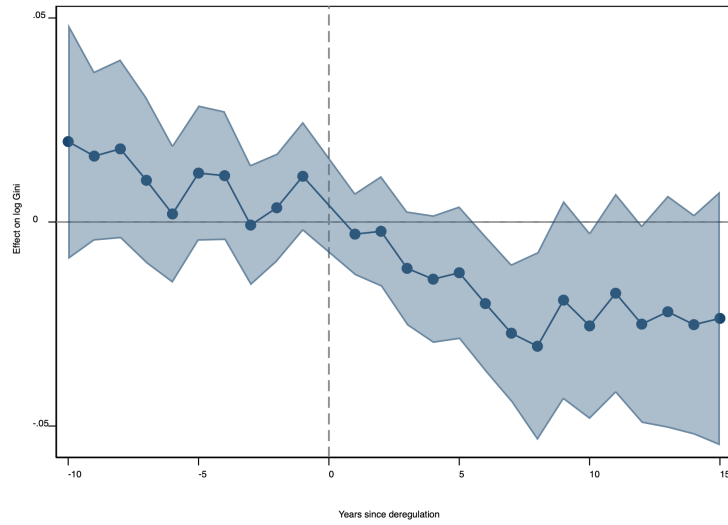


Figure 2: Original Beck et al. Event Study on Changes in Log Gini

Notes: This figure reproduces the two-way fixed effects event study from Beck, Levine, and Levkov (2010, Figure 3, Panel A). The vertical axis reports the estimated effect on the log Gini coefficient at each year relative to the onset of each branching deregulation, with year $t = -1$ omitted as the reference period. The horizontal axis reports years since deregulation, with endpoints at $t = -10$ and $t = 15$. Dots are point estimates from a TWFE regression of log Gini on a full set of event-time indicators, state fixed effects, and year fixed effects, estimated on the Beck et al. sample of 48 states plus the District of Columbia over 1976–2007 (omitting South Dakota and Delaware). The shaded region is the 95 percent confidence interval, constructed from standard errors clustered at the state level.

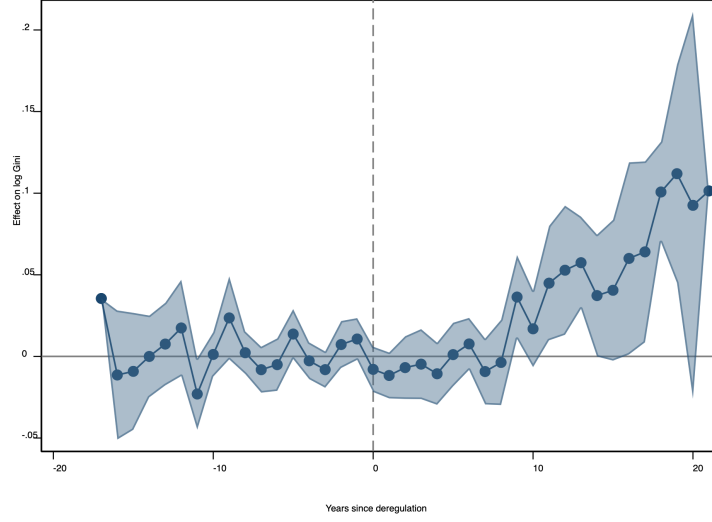


Figure 3: Callaway-Sant'Anna Event Study on Changes in Log Gini

Notes: This figure displays the dynamic treatment effects of branching deregulation on the log Gini coefficient, estimated using the Callaway and Sant'Anna (2021) staggered difference-in-differences estimator. The vertical axis reports the estimated effect on log Gini at each year relative to deregulation, with $t = -1$ as the reference period. The horizontal axis reports years since deregulation. Dots are event-time coefficients; the shaded regions represent the 95 percent confidence intervals for each point estimate. The 12 states that deregulated prior to 1976 are excluded from estimation as they lack any pre-treatment period.

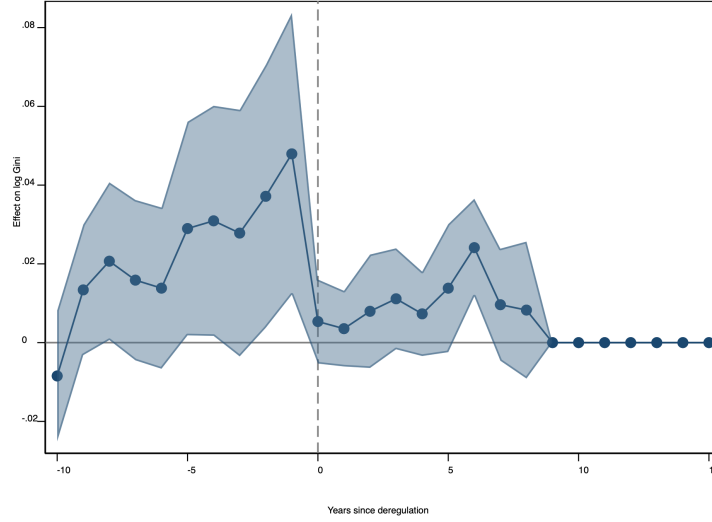


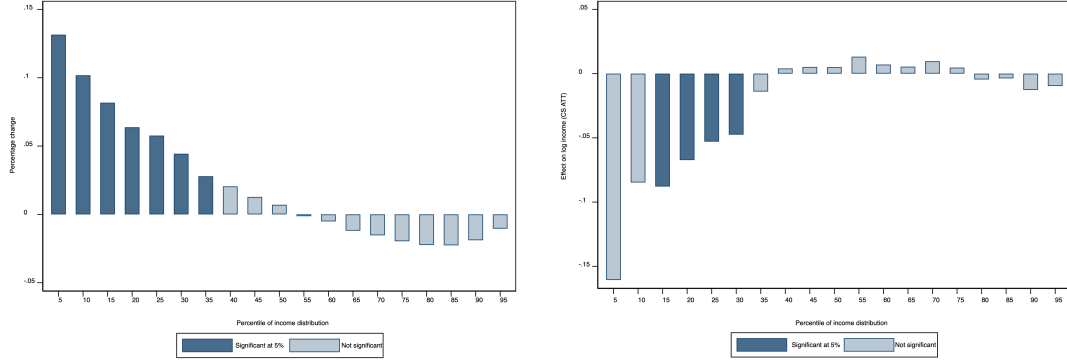
Figure 4: Borusyak & Jaravel & Spiess Event Study on Changes in Log Gini

Notes: This figure displays the dynamic treatment effects of intrastate branching deregulation on the log Gini coefficient, estimated using the Borusyak, Jaravel, and Spiess (2024) imputation estimator. The vertical axis reports the estimated effect on log Gini at each year relative to deregulation; the horizontal axis reports years since deregulation from $t = -10$ to $t = 15$. Dots are event-time coefficients; the shaded region are the 95 percent confidence intervals, with standard errors clustered at the state level. The 12 states that deregulated prior to 1976 are excluded because no untreated observations are available for them.

Table 2: Sensitivity of Estimated Effects to Estimation Window and Sample Composition

	Effect on Log Gini (1)	N (2)
<i>Panel A: Two-Way Fixed Effects</i>		
All 49 states, 1975–2006 (original Table II)	−0.022*** (0.007)	1,519
Dropping 12 always-treated states, 1975–2006	−0.017** (0.008)	1,147
<i>Panel B: Callaway and Sant’Anna</i>		
Baseline, 1975–2006	0.011 (0.008)	828
Restricted to 1975–1993 window	−0.015* (0.009)	578
Restricted to 1975–1990 window	−0.011 (0.011)	434

Notes: This table examines how the estimated effect of bank deregulation on the log Gini coefficient varies across estimation approaches and sample windows. Panel A reports two-way fixed effects estimates: the first row replicates the original specification using all 49 states; the second row drops the 12 states that deregulated before 1976 (always-treated units). Panel B reports Callaway and Sant’Anna (2021) estimates using not-yet-treated states as the control group. The first row in Panel B reports the baseline CS specification over the full sample period. The second and third rows restrict the estimation window to end in 1993 and 1990, respectively; cohorts that deregulated after the window endpoint are reassigned to the not-yet-treated control group. Standard errors in parentheses are clustered at the state level. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.



(a) Effects by Income Percentile in Beck et al. (b) Effects by Income Percentile under CSA

Figure 5: Effects by Percentile: Beck et al. TWFE vs. Callaway-Sant’Anna

Notes: This figure presents decomposed effects of intrastate branching deregulation on income inequality by each income percentile in the state-level income distribution, from the 5th to the 95th. Panel (a) reproduces Figure 2, Panel A of Beck, Levine, and Levkov (2010), estimated using the two-way fixed effects specification of their equation (1) on the full sample of 48 states plus the District of Columbia. Panel (b) reports the analogous coefficients obtained by re-estimating the same specification using the Callaway and Sant’Anna (2021) staggered difference-in-differences estimator with not-yet-treated states as the control group and the overall simple aggregation of group-time ATTs; the 12 states that deregulated prior to 1976 are excluded due to the lack of a pre-treatment period. Dark bars denote estimates statistically significant at the 5% level; light bars are not significant. In both panels the vertical axis reports a log-point effect, equivalent to a percentage change.

Table 3: Honest DiD Sensitivity Analysis of the Callaway-Sant’Anna Estimate

$\bar{M}$	95% Confidence Interval	
	Lower bound	Upper bound
	(1)	(2)
Original CS CI (parallel trends imposed)	0.007	0.033
$\bar{M} = 0.00$	0.007	0.033
$\bar{M} = 0.25$	-0.002	0.041
$\bar{M} = 0.50$	-0.012	0.050
$\bar{M} = 0.75$	-0.022	0.059
$\bar{M} = 1.00$	-0.033	0.069
$\bar{M} = 1.25$	-0.044	0.079
$\bar{M} = 1.50$	-0.055	0.090
$\bar{M} = 1.75$	-0.066	0.101
$\bar{M} = 2.00$	-0.077	0.111

Notes: This table reports Rambachan and Roth (2023) Honest DiD sensitivity bounds for the average post-treatment effect on the log Gini coefficient, computed from the Callaway and Sant’Anna (2021) event-study estimates over the full sample. The $\bar{M}$ parameter bounds the magnitude of post-treatment parallel-trends violations relative to the largest observed pre-treatment violation under the relative-magnitudes (Δ^{RM}) restriction. $\bar{M} = 0$ imposes exact parallel trends in the post-treatment period and reproduces the original CS confidence interval. $\bar{M} = 1$ allows post-treatment violations as large as the largest pre-treatment violation.

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